

Lesson 2 — Introduction to Character Displays

Exercise No. 1

Name: Jericho Francisco

Grade:

Course/Section: BSCPE 4B

Date: September 5, 2026

Machine Problem:

Create a program that will display the same output below using the hexadecimal value of its ASCII code and the carriage return and line feed option. Follow the given format with Uppercase and Lower case letters combination. Save your assembly file using Exer1.asm

First Name
Middle Initial
Last Name

Program:

(Save as Exer1.asm — paste your code screenshot below)

C: > Users > jeric > OneDrive > Documents > SIR CHARLES 4TH YR 1ST EM > **asm** EXER1.ASM > ...

```
1  .model small
2  .stack
3  .code
4  start:
5      mov ah,02h
6      mov dl,4ah
7      int 21h    ; 'J'
8      mov dl,45h
9      int 21h    ; 'E'
10     mov dl,52h
11     int 21h    ; 'R'
12     mov dl,49h
13     int 21h    ; 'I'
14     mov dl,43h
15     int 21h    ; 'C'
16     mov dl,48h
17     int 21h    ; 'H'
18     mov dl,4fh
19     int 21h    ; 'O'
20     mov dl,0dh
21     int 21h    ; carriage return
22     mov dl,0ah
23     int 21h    ; line feed
24     mov dl,4dh
25     int 21h    ; 'M'
26     mov dl,2eh
27     int 21h    ; '.'
28     mov dl,0dh
29     int 21h    ; carriage return
30     mov dl,0ah
31     int 21h    ; line feed
32     mov dl,46h
33     int 21h    ; 'F'
34     mov dl,52h
35     int 21h    ; 'R'
36     mov dl,41h
37     int 21h    ; 'A'
38     mov dl,4eh
39     int 21h    ; 'N'
40     mov dl,43h
41     int 21h    ; 'C'
42     mov dl,49h
43     int 21h    ; 'I'
44     mov dl,53h
45     int 21h    ; 'S'
46     mov dl,43h
47     int 21h    ; 'C'
```

```

47     inc 21h    ; C
48     mov dl,4fh
49     int 21h    ; 'O'
50     mov ah,4ch
51     int 21h
52 end start

```

Program Output:

(paste your output screenshot below)

```

Issue or suggestion: https://github.com/caiiycuk/js-dos/issues
HAVE FUN WITH js-dos/DOSBox-X !
Z:\>mount c ./c/
Drive C is mounted as local directory ./c/
Z:\>mount d ./d/
Drive D is mounted as local directory ./d/
Z:\>PATH Z:\;Z:\SYSTEM;Z:\BIN;Z:\DOS;Z:\4DOS;Z:\DEBUG;Z:\TEXTUTIL;C:\TASM
Z:\>d:
D:\>cd d:\
D:\>IASM D:\test.ASM
Turbo Assembler Version 4.1 Copyright (c) 1988, 1996 Borland International

Assembling file: D:\test.ASM to test.OBJ
Error messages: None
Warning messages: None
Passes: 1
Remaining memory: 441k

D:\>IF NOT ERRORLEVEL 1 TLINK D:\test
Turbo Link Version 7.1.30.1. Copyright (c) 1987, 1996 Borland International
D:\>IF NOT ERRORLEVEL 1 D:\test
JERICHO
M.
FRANCISCOD:\>[_

```

Lesson 2 — Introduction to Character Displays

Exercise No. 2

Name: Jericho Francisco

Grade:

Course/Section: BSCPE 4B

Date: September 5, 2026

Machine Problem:

Create a program that will display your personal information on the computer screen using hexadecimal value of ASCII Codes. Follow the given format below. Save the file using Exer2.asm as the file name.

```

      Name:  Juan M. dela Cruz
      Birth Date: August 4, 1989
      e-mail address: juan_delacruz@mail.com
Address: 2347-B Memory St. Processor Village Calookan Manila
      Province: Infanta Quezon
      Contact Number: 02-7443567
Guardian's Name: Sixto C. dela Cruz
      Occupation: Government Employee

```

Program:

(Save as Exer2.asm — paste your code screenshot below)

```
1  .model small
2  .stack
3  .code
4  start:
5      mov ah,02h
6      mov dl,4eh
7      int 21h ; 'N'
8      mov dl,61h
9      int 21h ; 'a'
10     mov dl,6dh
11     int 21h ; 'm'
12     mov dl,65h
13     int 21h ; 'e'
14     mov dl,3ah
15     int 21h ; ':'
16     mov dl,20h
17     int 21h ; space
18     mov dl,4ah
19     int 21h ; 'J'
20     mov dl,65h
21     int 21h ; 'e'
22     mov dl,72h
23     int 21h ; 'r'
24     mov dl,69h
25     int 21h ; 'i'
26     mov dl,63h
27     int 21h ; 'c'
28     mov dl,68h
29     int 21h ; 'h'
30     mov dl,6fh
31     int 21h ; 'o'
32     mov dl,20h
33     int 21h ; space
34     mov dl,4dh
35     int 21h ; 'M'
36     mov dl,2eh
37     int 21h ; '.'
38     mov dl,20h
39     int 21h ; space
40     mov dl,46h
41     int 21h ; 'F'
42     mov dl,72h
43     int 21h ; 'r'
44     mov dl,61h
45     int 21h ; 'a'
46     mov dl,6eh
47     int 21h ; 'n'
```

```
48     mov dl,63h
49     int 21h    ; 'c'
50     mov dl,69h
51     int 21h    ; 'i'
52     mov dl,73h
53     int 21h    ; 's'
54     mov dl,63h
55     int 21h    ; 'c'
56     mov dl,6fh
57     int 21h    ; 'o'
58     mov dl,0dh
59     int 21h    ; carriage return
60     mov dl,0ah
61     int 21h    ; line feed
62     mov dl,42h
63     int 21h    ; 'B'
64     mov dl,69h
65     int 21h    ; 'i'
66     mov dl,72h
67     int 21h    ; 'r'
68     mov dl,74h
69     int 21h    ; 't'
70     mov dl,68h
71     int 21h    ; 'h'
72     mov dl,20h
73     int 21h    ; space
74     mov dl,44h
75     int 21h    ; 'D'
76     mov dl,61h
77     int 21h    ; 'a'
78     mov dl,74h
79     int 21h    ; 't'
80     mov dl,65h
81     int 21h    ; 'e'
82     mov dl,3ah
83     int 21h    ; ':'
84     mov dl,20h
85     int 21h    ; space
86     mov dl,44h
87     int 21h    ; 'D'
88     mov dl,65h
89     int 21h    ; 'e'
90     mov dl,63h
91     int 21h    ; 'c'
92     mov dl,65h
93     int 21h    ; 'e'
94     mov dl,6dh
```

```
95     int 21h    ; 'm'
96     mov dl,62h
97     int 21h    ; 'b'
98     mov dl,65h
99     int 21h    ; 'e'
100    mov dl,72h
101    int 21h    ; 'r'
102    mov dl,20h
103    int 21h    ; space
104    mov dl,31h
105    int 21h    ; '1'
106    mov dl,36h
107    int 21h    ; '6'
108    mov dl,2ch
109    int 21h    ; ','
110    mov dl,20h
111    int 21h    ; space
112    mov dl,32h
113    int 21h    ; '2'
114    mov dl,30h
115    int 21h    ; '0'
116    mov dl,30h
117    int 21h    ; '0'
118    mov dl,34h
119    int 21h    ; '4'
120    mov dl,0dh
121    int 21h    ; carriage return
122    mov dl,0ah
123    int 21h    ; line feed
124    mov dl,65h
125    int 21h    ; 'e'
126    mov dl,2dh
127    int 21h    ; '-'
128    mov dl,6dh
129    int 21h    ; 'm'
130    mov dl,61h
131    int 21h    ; 'a'
132    mov dl,69h
133    int 21h    ; 'i'
134    mov dl,6ch
135    int 21h    ; 'l'
136    mov dl,20h
137    int 21h    ; space
138    mov dl,61h
139    int 21h    ; 'a'
140    mov dl,64h
141    int 21h    ; 'd'
```

```
142     mov dl,64h
143     int 21h    ; 'd'
144     mov dl,72h
145     int 21h    ; 'r'
146     mov dl,65h
147     int 21h    ; 'e'
148     mov dl,73h
149     int 21h    ; 's'
150     mov dl,73h
151     int 21h    ; 's'
152     mov dl,3ah
153     int 21h    ; ':'
154     mov dl,20h
155     int 21h    ; space
156     mov dl,6ah
157     int 21h    ; 'j'
158     mov dl,65h
159     int 21h    ; 'e'
160     mov dl,72h
161     int 21h    ; 'r'
162     mov dl,69h
163     int 21h    ; 'i'
164     mov dl,63h
165     int 21h    ; 'c'
166     mov dl,68h
167     int 21h    ; 'h'
168     mov dl,6fh
169     int 21h    ; 'o'
170     mov dl,6dh
171     int 21h    ; 'm'
172     mov dl,61h
173     int 21h    ; 'a'
174     mov dl,62h
175     int 21h    ; 'b'
176     mov dl,61h
177     int 21h    ; 'a'
178     mov dl,6eh
179     int 21h    ; 'n'
180     mov dl,65h
181     int 21h    ; 'e'
182     mov dl,73h
183     int 21h    ; 's'
184     mov dl,66h
185     int 21h    ; 'f'
186     mov dl,72h
187     int 21h    ; 'r'
188     mov dl,61h
```

C:\Users\jeric\OneDrive\Documents\SIR CHARLES 4TH YR

```
190     mov dl,6eh
191     int 21h    ; 'n'
192     mov dl,63h
193     int 21h    ; 'c'
194     mov dl,69h
195     int 21h    ; 'i'
196     mov dl,73h
197     int 21h    ; 's'
198     mov dl,63h
199     int 21h    ; 'c'
200     mov dl,6fh
201     int 21h    ; 'o'
202     mov dl,40h
203     int 21h    ; '@'
204     mov dl,67h
205     int 21h    ; 'g'
206     mov dl,6dh
207     int 21h    ; 'm'
208     mov dl,61h
209     int 21h    ; 'a'
210     mov dl,69h
211     int 21h    ; 'i'
212     mov dl,6ch
213     int 21h    ; 'l'
214     mov dl,2eh
215     int 21h    ; '.'
216     mov dl,63h
217     int 21h    ; 'c'
218     mov dl,6fh
219     int 21h    ; 'o'
220     mov dl,6dh
221     int 21h    ; 'm'
222     mov dl,0dh
223     int 21h    ; carriage return
224     mov dl,0ah
225     int 21h    ; line feed
226     mov dl,41h
227     int 21h    ; 'A'
228     mov dl,64h
229     int 21h    ; 'd'
230     mov dl,64h
231     int 21h    ; 'd'
232     mov dl,72h
233     int 21h    ; 'r'
234     mov dl,65h
235     int 21h    ; 'e'
236     mov dl,73h
```

```
C: > Users > jeric > OneDrive > Documents > SIR CHARLES 4TH YR 1ST
236      mov dl,73h
237      int 21h      ; 's'
238      mov dl,73h
239      int 21h      ; 's'
240      mov dl,3ah
241      int 21h      ; ':'
242      mov dl,20h
243      int 21h      ; space
244      mov dl,53h
245      int 21h      ; 'S'
246      mov dl,69h
247      int 21h      ; 'i'
248      mov dl,6ch
249      int 21h      ; 'l'
250      mov dl,61h
251      int 21h      ; 'a'
252      mov dl,6eh
253      int 21h      ; 'n'
254      mov dl,67h
255      int 21h      ; 'g'
256      mov dl,61h
257      int 21h      ; 'a'
258      mov dl,6eh
259      int 21h      ; 'n'
260      mov dl,20h
261      int 21h      ; space
262      mov dl,49h
263      int 21h      ; 'I'
264      mov dl,6ch
265      int 21h      ; 'l'
266      mov dl,61h
267      int 21h      ; 'a'
268      mov dl,79h
269      int 21h      ; 'y'
270      mov dl,61h
271      int 21h      ; 'a'
272      mov dl,2ch
273      int 21h      ; ','
274      mov dl,20h
275      int 21h      ; space
276      mov dl,44h
277      int 21h      ; 'D'
278      mov dl,61h
279      int 21h      ; 'a'
280      mov dl,72h
281      int 21h      ; 'r'
282      mov dl,61h
283      int 21h      ; 'a'
```

```
284     mov dl,73h
285     int 21h     ; 's'
286     mov dl,61h
287     int 21h     ; 'a'
288     mov dl,2ch
289     int 21h     ; ','
290     mov dl,20h
291     int 21h     ; space
292     mov dl,54h
293     int 21h     ; 'T'
294     mov dl,61h
295     int 21h     ; 'a'
296     mov dl,6eh
297     int 21h     ; 'n'
298     mov dl,61h
299     int 21h     ; 'a'
300     mov dl,75h
301     int 21h     ; 'u'
302     mov dl,61h
303     int 21h     ; 'a'
304     mov dl,6eh
305     int 21h     ; 'n'
306     mov dl,20h
307     int 21h     ; space
308     mov dl,43h
309     int 21h     ; 'C'
310     mov dl,69h
311     int 21h     ; 'i'
312     mov dl,74h
313     int 21h     ; 't'
314     mov dl,79h
315     int 21h     ; 'y'
316     mov dl,0dh
317     int 21h     ; carriage return
318     mov dl,0ah
319     int 21h     ; line feed
320     mov dl,50h
321     int 21h     ; 'P'
322     mov dl,72h
323     int 21h     ; 'r'
324     mov dl,6fh
325     int 21h     ; 'o'
326     mov dl,76h
327     int 21h     ; 'v'
328     mov dl,69h
329     int 21h     ; 'i'
330     mov dl,6eh
```

C: > Users > jeric > OneDrive > Documents > SIR CHARLES 4TH YR 1ST EM > ASM EXER2.A

```
331     int 21h    ; 'n'
332     mov dl,63h
333     int 21h    ; 'c'
334     mov dl,65h
335     int 21h    ; 'e'
336     mov dl,3ah
337     int 21h    ; ':'
338     mov dl,20h
339     int 21h    ; space
340     mov dl,42h
341     int 21h    ; 'B'
342     mov dl,61h
343     int 21h    ; 'a'
344     mov dl,74h
345     int 21h    ; 't'
346     mov dl,61h
347     int 21h    ; 'a'
348     mov dl,6eh
349     int 21h    ; 'n'
350     mov dl,67h
351     int 21h    ; 'g'
352     mov dl,61h
353     int 21h    ; 'a'
354     mov dl,73h
355     int 21h    ; 's'
356     mov dl,0dh
357     int 21h    ; carriage return
358     mov dl,0ah
359     int 21h    ; line feed
360     mov dl,43h
361     int 21h    ; 'C'
362     mov dl,6fh
363     int 21h    ; 'o'
364     mov dl,6eh
365     int 21h    ; 'n'
366     mov dl,74h
367     int 21h    ; 't'
368     mov dl,61h
369     int 21h    ; 'a'
370     mov dl,63h
371     int 21h    ; 'c'
372     mov dl,74h
373     int 21h    ; 't'
374     mov dl,20h
375     int 21h    ; space
376     mov dl,4eh
377     int 21h    ; 'N'
```

```

377     int 21h      ; "
378     mov dl,75h
379     int 21h      ; 'u'
380     mov dl,6dh
381     int 21h      ; 'm'
382     mov dl,62h
383     int 21h      ; 'b'
384     mov dl,65h
385     int 21h      ; 'e'
386     mov dl,72h
387     int 21h      ; 'r'
388     mov dl,3ah
389     int 21h      ; ':'
390     mov dl,20h
391     int 21h      ; space
392     mov dl,30h
393     int 21h      ; '0'
394     mov dl,39h
395     int 21h      ; '9'
396     mov dl,39h
397     int 21h      ; '9'
398     mov dl,34h
399     int 21h      ; '4'
400     mov dl,31h
401     int 21h      ; '1'
402     mov dl,32h
403     int 21h      ; '2'
404     mov dl,35h
405     int 21h      ; '5'
406     mov dl,37h
407     int 21h      ; '7'
408     mov dl,31h
409     int 21h      ; '1'
410     mov dl,31h
411     int 21h      ; '1'
412     mov dl,32h
413     int 21h      ; '2'
414     mov dl,0dh
415     int 21h      ; carriage return
416     mov dl,0ah
417     int 21h      ; line feed
418     mov dl,47h
419     int 21h      ; 'G'
420     mov dl,75h
421     int 21h      ; 'u'
422     mov dl,61h
423     int 21h      ; 'a'
424     mov dl,72h

```

C:\> Users > jeric > OneDrive > Documents > SIR CHARLES 4TH YR 1ST EM > ASM EX

```
425     int 21h    ; 'r'
426     mov dl,64h
427     int 21h    ; 'd'
428     mov dl,69h
429     int 21h    ; 'i'
430     mov dl,61h
431     int 21h    ; 'a'
432     mov dl,6eh
433     int 21h    ; 'n'
434     mov dl,27h
435     int 21h    ; apostrophe
436     mov dl,73h
437     int 21h    ; 's'
438     mov dl,20h
439     int 21h    ; space
440     mov dl,4eh
441     int 21h    ; 'N'
442     mov dl,61h
443     int 21h    ; 'a'
444     mov dl,6dh
445     int 21h    ; 'm'
446     mov dl,65h
447     int 21h    ; 'e'
448     mov dl,3ah
449     int 21h    ; ':'
450     mov dl,20h
451     int 21h    ; space
452     mov dl,47h
453     int 21h    ; 'G'
454     mov dl,69h
455     int 21h    ; 'i'
456     mov dl,72h
457     int 21h    ; 'r'
458     mov dl,6ch
459     int 21h    ; 'l'
460     mov dl,69h
461     int 21h    ; 'i'
462     mov dl,65h
463     int 21h    ; 'e'
464     mov dl,20h
465     int 21h    ; space
466     mov dl,4dh
467     int 21h    ; 'M'
468     mov dl,2eh
469     int 21h    ; '.'
470     mov dl,20h
471     int 21h    ; space
```

C: > Users > jeric > OneDrive > Documents > SIR CHARLES 4TH YR 1ST

```
472     mov dl,46h
473     int 21h    ; 'F'
474     mov dl,72h
475     int 21h    ; 'r'
476     mov dl,61h
477     int 21h    ; 'a'
478     mov dl,6eh
479     int 21h    ; 'n'
480     mov dl,63h
481     int 21h    ; 'c'
482     mov dl,69h
483     int 21h    ; 'i'
484     mov dl,73h
485     int 21h    ; 's'
486     mov dl,63h
487     int 21h    ; 'c'
488     mov dl,6fh
489     int 21h    ; 'o'
490     mov dl,0dh
491     int 21h    ; carriage return
492     mov dl,0ah
493     int 21h    ; line feed
494     mov dl,4fh
495     int 21h    ; 'O'
496     mov dl,63h
497     int 21h    ; 'c'
498     mov dl,63h
499     int 21h    ; 'c'
500     mov dl,75h
501     int 21h    ; 'u'
502     mov dl,70h
503     int 21h    ; 'p'
504     mov dl,61h
505     int 21h    ; 'a'
506     mov dl,74h
507     int 21h    ; 't'
508     mov dl,69h
509     int 21h    ; 'i'
510     mov dl,6fh
511     int 21h    ; 'o'
512     mov dl,6eh
513     int 21h    ; 'n'
514     mov dl,3ah
515     int 21h    ; ':'
516     mov dl,20h
517     int 21h    ; space
518     mov dl,41h
519     int 21h    ; 'A'
```

```
519     int 21h    ; 'A'
520     mov dl,73h
521     int 21h    ; 's'
522     mov dl,73h
523     int 21h    ; 's'
524     mov dl,69h
525     int 21h    ; 'i'
526     mov dl,73h
527     int 21h    ; 's'
528     mov dl,74h
529     int 21h    ; 't'
530     mov dl,61h
531     int 21h    ; 'a'
532     mov dl,6eh
533     int 21h    ; 'n'
534     mov dl,74h
535     int 21h    ; 't'
536     mov dl,20h
537     int 21h    ; space
538     mov dl,4ch
539     int 21h    ; 'L'
540     mov dl,65h
541     int 21h    ; 'e'
542     mov dl,61h
543     int 21h    ; 'a'
544     mov dl,64h
545     int 21h    ; 'd'
546     mov dl,65h
547     int 21h    ; 'e'
548     mov dl,72h
549     int 21h    ; 'r'
550     mov dl,20h
551     int 21h    ; space
552     mov dl,2fh
553     int 21h    ; '/'
554     mov dl,20h
555     int 21h    ; space
556     mov dl,46h
557     int 21h    ; 'F'
558     mov dl,61h
559     int 21h    ; 'a'
560     mov dl,63h
561     int 21h    ; 'c'
562     mov dl,74h
563     int 21h    ; 't'
564     mov dl,6fh
565     int 21h    ; 'o'
```

```
C: > Users > jeric > OneDrive > Documents > SIR CHARLES 4
566      mov dl,72h
567      int 21h    ; 'r'
568      mov dl,79h
569      int 21h    ; 'y'
570      mov dl,20h
571      int 21h    ; space
572      mov dl,57h
573      int 21h    ; 'W'
574      mov dl,6fh
575      int 21h    ; 'o'
576      mov dl,72h
577      int 21h    ; 'r'
578      mov dl,6bh
579      int 21h    ; 'k'
580      mov dl,65h
581      int 21h    ; 'e'
582      mov dl,72h
583      int 21h    ; 'r'
584      mov dl,0dh
585      int 21h    ; carriage return
586      mov dl,0ah
587      int 21h    ; line feed
588      mov ah,4ch
589      int 21h
590      end start
```

Program Output:

(paste your output screenshot below)

```
Drive D is mounted as local directory ./d/
Z:\>PATH Z:\Z:\SYSTEM\Z:\BIN\Z:\DOS\Z:\4DOS\Z:\DEBUG\Z:\TEXTUTIL\C:\TASM
Z:\>d:
D:\>cd d:\
D:\>TASM D:\test.ASM
Turbo Assembler Version 4.1 Copyright (c) 1988, 1996 Borland International
```

```
Assembling file:  D:\test.ASM to test.OBJ
Error messages:   None
Warning messages: None
Passes:           1
Remaining memory: 439k
```

```
D:\>IF NOT ERRORLEVEL 1 TLINK D:\test
Turbo Link Version 7.1.30.1. Copyright (c) 1987, 1996 Borland International
D:\>IF NOT ERRORLEVEL 1 D:\test
Name: Jericho M. Francisco
Birth Date: December 16, 2004
e-mail address: jerichonabanesfrancisco@gmail.com
Address: Silangan Ilaya, Darasa, Tanauan City
Province: Batangas
Contact Number: 09941257112
Guardian's Name: Girlie M. Francisco
Occupation: Assistant Leader / Factory Worker
D:\>
```

Grade:

Date: September 5, 2026

Create a program that will display your first name and surname in the computer screen using hexadecimal value of ASCII Codes. Other character may also be used for the frame. Follow the given format below. Save your program using Exer3.asm as the filename.

(Save as Exer3.asm — paste your code screenshot below)

[illegible]

```
int 21h  ; '@'
mov dl,40h
int 21h  ; '@'
mov dl,40h
int 21h  ; '@'
mov dl,40h
int 21h  ; '@'
mov dl,40h
int 21h  ; '@'
mov dl,40h
int 21h  ; '@'
mov dl,0dh
int 21h  ; carriage return
mov dl,0ah
int 21h  ; line feed
mov dl,40h
int 21h  ; '@'
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
mov dl,40h
int 21h  ; '@'
mov dl,0dh
int 21h  ; carriage return
mov dl,0ah
int 21h  ; line feed
mov dl,40h
int 21h  ; '@'
mov dl,20h
int 21h  ; space
mov dl,20h
int 21h  ; space
```

```

mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,40h
int 21h ; '@'
mov dl,0dh
int 21h ; carriage return
mov dl,0ah
int 21h ; line feed
mov dl,40h
int 21h ; '@'
mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,4ah
int 21h ; 'J'
mov dl,45h
int 21h ; 'E'
mov dl,52h
int 21h ; 'R'
mov dl,49h
int 21h ; 'T'
mov dl,43h
int 21h ; 'C'
mov dl,48h
int 21h ; 'H'
mov dl,4fh
int 21h ; 'O'
mov dl,20h
int 21h ; space
mov dl,20h
int 21h ; space
mov dl,20h

```

```

int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,40h
int 21h    ; '@'
mov dl,0dh
int 21h    ; carriage return
mov dl,0ah
int 21h    ; line feed
mov dl,40h
int 21h    ; '@'
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,40h
int 21h    ; '@'
mov dl,0dh
int 21h    ; carriage return
mov dl,0ah
int 21h    ; line feed
mov dl,40h
int 21h    ; '@'
mov dl,20h
int 21h    ; space
mov dl,20h
int 21h    ; space
mov dl,46h
int 21h    ; 'F'
mov dl,52h
int 21h    ; 'R'

```

[illegible]

[illegible]

```
mov dl,40h
int 21h ; '@'
mov dl,40h
int 21h ; '@'
mov dl,40h
int 21h ; '@'
mov dl,40h
int 21h ; '@'
mov dl,40h
int 21h ; '@'
mov dl,40h
int 21h ; '@'
mov dl,40h
int 21h ; '@'
mov dl,40h
int 21h ; '@'
mov dl,40h
int 21h ; '@'
mov dl,0dh
int 21h ; carriage return
mov dl,0ah
int 21h ; line feed
mov ah,4ch
int 21h
end start
```

Program Output:

(paste your output screenshot below)

Lesson 3 — Program Looping

Exercise No. 4

Name: Jericho Francisco

Grade:

Course/Section: BSCPE 4B

Date: September 5, 2026

Machine Problem:

Create a program that will display the output format shown below. Save the program using Exer4.asm as the filename.

AaBaCcDdEeFf Zz

Program:

(Save as Exer4.asm — paste your code screenshot below)

```
1  .model small
2  .stack
3  .code
4  start:
5      mov ah, 02h
6      mov cx, 26      ; A to Z = 26 letters
7      mov bl, 41h     ; uppercase counter, starts at 'A'
8      mov bh, 61h     ; lowercase counter, starts at 'a'
9  ulit:
10     mov dl, bl
11     int 21h          ; display uppercase letter
12     mov dl, bh
13     int 21h          ; display lowercase letter
14     inc bl           ; advance uppercase counter
15     inc bh           ; advance lowercase counter
16     loop ulit
17     mov ah, 4ch
18     int 21h
19  end start
```

Program Output:

(paste your output screenshot below)

Homepage of project: https://jsdos.com/	Complete DOS emulations
DOSBox-X guides: https://dosbox-x.com/wiki	DOS, Windows 3.x and 9x
Issue or suggestion: https://github.com/caiiycuk/js-dos/issues	

HAVE FUN WITH js-dos/DOSBox-X !

Z:\>mount c ./c/

Drive C is mounted as local directory ./c/

Z:\>mount d ./d/

Drive D is mounted as local directory ./d/

Z:\>PATH Z:\;Z:\SYSTEM;Z:\BIN;Z:\DOS;Z:\4DOS;Z:\DEBUG;Z:\TEXTUTIL;C:\TASM

Z:\>d:

D:\>cd d:\

D:\>TASM D:\test.ASM

Turbo Assembler Version 4.1 Copyright (c) 1988, 1996 Borland International

Assembling file: D:\test.ASM to test.OBJ

Error messages: None

Warning messages: None

Passes: 1

Remaining memory: 442k

D:\>IF NOT ERRORLEVEL 1 TLINK D:\test

Turbo Link Version 7.1.30.1. Copyright (c) 1987, 1996 Borland International

D:\>IF NOT ERRORLEVEL 1 D:\test

AaBbCcDdEeFfGgHhIiJjKkLlMmNnOoPpQqRrSsTtUuVvWwXxYyZzD:\>_

Lesson 3 — Program Looping

Exercise No. 5

Name: Jericho Francisco

Grade:

Course/Section: BSCPE 4B

Date: September 5, 2026

Machine Problem:

Create a program that will display the output format shown below. Save the program using Exer5.asm as the filename.

zAyBxCwDu..... aZ

Program:

(Save as Exer5.asm — paste your code screenshot below)

```
1  .model small
2  .stack
3  .code
4  start:
5      mov ah, 02h
6      mov cx, 26      ; z down to a = 26 letters
7      mov bl, 7ah      ; lowercase counter, starts at 'z' (descending)
8      mov bh, 41h      ; uppercase counter, starts at 'A' (ascending)
9  ulit:
10     mov dl, bl
11     int 21h          ; display lowercase letter
12     mov dl, bh
13     int 21h          ; display uppercase letter
14     dec bl           ; move lowercase letter backward (z,y,x,...)
15     inc bh           ; move uppercase letter forward (A,B,C,...)
16     loop ulit
17     mov ah, 4ch
18     int 21h
19  end start
```

Program Output:

(paste your output screenshot below)

Homepage of project: <https://jsdos.com/> Complete DOS emulations
DOSBox-X guides: <https://dosbox-x.com/wiki> DOS, Windows 3.x and 9x
Issue or suggestion: <https://github.com/caiiycuk/js-dos/issues>

HAVE FUN WITH js-dos/DOSBox-X !

Z:\>mount c ./c/

Drive C is mounted as local directory ./c/

Z:\>mount d ./d/

Drive D is mounted as local directory ./d/

Z:\>PATH Z:\;Z:\SYSTEM;Z:\BIN;Z:\DOS;Z:\4DOS;Z:\DEBUG;Z:\TEXTUTIL;C:\TASM

Z:\>d:

D:\>cd d:\

D:\>TASM D:\test.ASM

Turbo Assembler Version 4.1 Copyright (c) 1988, 1996 Borland International

Assembling file: D:\test.ASM to test.OBJ

Error messages: None

Warning messages: None

Passes: 1

Remaining memory: 442k

D:\>IF NOT ERRORLEVEL 1 TLINK D:\test

Turbo Link Version 7.1.30.1. Copyright (c) 1987, 1996 Borland International

D:\>IF NOT ERRORLEVEL 1 D:\test

zAyBxCwDvEuFtGsHrIqJpKoLnMmNlOkPjQiRhSgTfUeVdWcXbYaZD:\>

Exercise No. 6

Name: Jericho Francisco

Grade:

Course/Section: BSCPE 4B

Date: September 5, 2026

Machine Problem:

Create a program that will display the output format shown below. Save the program using Exer6.asm as the filename.

```
aZ
bY
cX
dW
.
.
.
zA
```

Program:

(Save as Exer6.asm — paste your code screenshot below)

```
1  .model small
2  .stack
3  .code
4  start:
5      mov ah, 02h
6      mov cx, 26      ; a to z = 26 lines
7      mov bl, 61h      ; lowercase counter, starts at 'a' (ascending)
8      mov bh, 5ah      ; uppercase counter, starts at 'Z' (descending)
9  ulit:
10     mov dl, bl
11     int 21h          ; display lowercase letter
12     mov dl, bh
13     int 21h          ; display uppercase letter
14     mov dl, 0dh
15     int 21h          ; carriage return
16     mov dl, 0ah
17     int 21h          ; line feed
18     inc bl           ; move lowercase letter forward (a,b,c,...)
19     dec bh           ; move uppercase letter backward (Z,Y,X,...)
20     loop ulit
21     mov ah, 4ch
22     int 21h
23  end start
```

Program Output:

(paste your output screenshot below)

```
cX
dW
eU
fU
gT
hS
iR
jQ
kP
lO
mN
nM
oL
pK
qJ
rI
sH
tG
uF
vE
wD
xC
yB
zA
D:\>
```

Lesson 3 — Program Looping

Exercise No. 7

Name: Jericho Francisco

Grade:

Course/Section: BSCPE 4B

Date: September 5, 2026

Machine Problem:

Create a program that will display the output format shown below. Save the program using Exer7.asm as the filename.

Az
By
Cx
Dw
Ev
Fu
.
.
.
Za

Program:

(Save as Exer7.asm — paste your code screenshot below)

```

1  .model small
2  .stack
3  .code
4  start:
5      mov ah, 02h
6      mov cx, 26      ; A to Z = 26 lines
7      mov bl, 41h     ; uppercase counter, starts at 'A' (ascending)
8      mov bh, 7ah     ; lowercase counter, starts at 'z' (descending)
9  ulit:
10     mov dl, bl
11     int 21h          ; display uppercase letter
12     mov dl, bh
13     int 21h          ; display lowercase letter
14     mov dl, 0dh
15     int 21h          ; carriage return
16     mov dl, 0ah
17     int 21h          ; line feed
18     inc bl           ; move uppercase letter forward (A,B,C,...)
19     dec bh           ; move lowercase letter backward (z,y,x,...)
20     loop ulit
21     mov ah, 4ch
22     int 21h
23 end start

```

Program Output:

(paste your output screenshot below)

Cx
Dw
Ev
Fu
Gt
Hs
Ir
Jq
Kp
Lo
Mn
Nm
Ol
Pk
Qj
Ri
Sh
Tg
Uf
Ve
Wd
Xc
Yb
Za
D:\>_